# Summary Table of Reviewed Papers

Distributed Deep Learning Training on GPU/TPU Clusters — HPC ISE-II | Aditi Mittal | PRN 23610067

| Sr. | Authors / Year | Title | Problem Addressed | Technique | Platform / Tools | Key Results | Limitations |
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| **1** | Huang et al., 2019 | GPipe | Training models too large for a single device | Synchronous pipeline parallelism with microbatches | TPU v3, TensorFlow | Trained 6B-param Transformer; feasible large-model training | Pipeline bubble; high activation memory |
| **2** | Harlap et al., 2019 | PipeDream | Communication bottleneck and throughput limits in data parallelism | Asynchronous pipeline with automatic layer partitioning | GPU cluster, PyTorch | Up to 5× faster time-to-accuracy; up to 95% less communication | Asynchronous updates complicate convergence; versioned weights add memory |
| **3** | Shoeybi et al., 2019 | Megatron-LM | Scaling transformer models beyond single-GPU memory | Tensor (model) parallelism within layers | NVIDIA DGX-2, 512 GPUs | 8.3B-param model; 76% scaling efficiency | Communication-intensive; limited to fast intra-node links |
| **4** | Rajbhandari et al., 2020 | ZeRO | Redundant optimizer state / gradient storage in data parallelism | Partition optimizer states, gradients, and parameters across workers | NVIDIA GPU cluster, DeepSpeed | >100B-param models on 400 GPUs; 15 petaflops | Communication overhead for parameter gathering; interconnect dependent |
| **5** | Fan et al., 2021 | DAPPLE | Stage imbalance and memory use in synchronous pipelines | Combined synchronous data and pipeline parallelism with automatic planner | NVIDIA GPU cluster | 1.6× throughput over GPipe; 12% lower memory | Planner accuracy depends on hardware cost model |
| **6** | Li et al., 2020 | PyTorch DDP | Communication overhead in synchronous data parallelism | Gradient bucketing with backward-communication overlap | NVIDIA GPU cluster, PyTorch | Near-linear scaling to 256 GPUs | Scaling is workload- and configuration-specific |
| **7** | Narayanan et al., 2021 | Megatron-LM (SC'21) | Scaling to trillion-parameter models on large GPU clusters | Tensor + pipeline + data parallelism (3D hybrid) | 3,072 NVIDIA A100 GPUs | 1T-param training; 502 petaflops; 52% of peak FLOPs | Requires large, specialized cluster; 48% FLOPs still idle |
| **8** | Rajbhandari et al., 2021 | ZeRO-Infinity | GPU memory wall limiting trainable model size | Heterogeneous memory offload (GPU + CPU + NVMe) via DeepSpeed | 512 NVIDIA V100 GPUs, NVMe | >25 petaflops; 1T-param fine-tuning on single DGX-2 | Offload bandwidth sensitive; storage I/O can be bottleneck |

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| **9** | Athlur et al., 2021 | Varuna | High cost of large-model training on dedicated clusters | Pipeline parallelism on low-priority cloud VMs with automatic configuration | AWS spot VMs, commodity networking | 200B-param model; up to 18× over model-parallel baseline | Results depend on spot pricing, interruptions, and fault handling |
| **10** | Li & Hoefler, 2021 | Chimera | Pipeline bubble overhead in synchronous pipelines | Bidirectional pipelines to reduce idle stages | NVIDIA GPU cluster | 1.16×–2.34× over single-direction pipeline baselines | Scheduling and placement complexity; requires suitable pipeline structure |
| **11** | Narayanan et al., 2021 | PipeDream-2BW | Activation and weight memory overhead in pipeline parallelism | Weight double buffering + gradient coalescing + automatic partitioning | NVIDIA GPU cluster | Up to 20× acceleration vs. baselines for GPT/BERT training | Large speedup claim is baseline-specific; details matter for reproduction |
| **12** | Zheng et al., 2022 | Alpa | Manual effort in designing parallelism strategies | Hierarchical compiler passes for inter- and intra-operator parallelism | NVIDIA GPU cluster, JAX | Matches or beats hand-tuned plans; supports heterogeneous hardware | Cost-model accuracy; limited support for dynamic/irregular models |
| **13** | Korthikanti et al., 2022 | Sequence Parallelism | High activation memory from full recomputation in Transformers | Sequence parallelism + selective activation recomputation | 2,240 NVIDIA A100 GPUs | 54.2% MFU vs. 42.1%; 5× activation memory reduction | Specific to large Transformer workloads on large clusters |
| **14** | Li et al., 2021 | TeraPipe | Coarse pipeline granularity for autoregressive models | Token-level pipeline parallelism with dynamic programming schedule | 48× AWS p3.16xlarge, PyTorch | Up to 5× speedup for 175B GPT-3 vs. model-parallel baseline | Requires autoregressive structure; not general to all model types |
| **15** | Lepikhin et al., 2021 | GShard | Scaling beyond dense model capacity limits | Sparse Mixture-of-Experts + compiler-based automatic sharding | 2,048 TPU v3 accelerators | 600B+ param multilingual model trained in 4 days | TPU-specific infrastructure; routing imbalance in sparse models |